

MODULE 4

1. Describe the architecture and components of a device controller. Explain the role of control registers, status registers, and data registers in I/O operations.
2. Explain the concept of I/O devices and describe the two main approaches for accessing I/O devices. Compare programmed I/O with interrupted I/O in terms of implementation, advantages, and disadvantages.
3. A system is designed to transfer a block of 1024 bytes from an external device to memory. Justify which data transfer technique (Programmed I/O, Interrupt-driven I/O, or DMA) would be most efficient for this scenario, and explain why.
4. Explain the three main I/O data transfer techniques. Describe the basic mechanism of each technique and identify their primary characteristics.
5. A CPU operates at 1 GHz and takes 50 cycles to handle one I/O interrupt. If an I/O device generates 5000 interrupts per second, calculate the percentage of CPU time spent in servicing I/O. Suggest a better technique to improve performance.
6. With the help of a neat diagram, explain the concept of cycle stealing in Direct Memory Access (DMA). How does this mechanism allow DMA to transfer data while sharing the system bus with the CPU? Analyze its impact on overall CPU performance.
7. A system needs to handle keyboard input, disk transfers, and network communication. Analyze which I/O transfer technique would be most appropriate for each device and justify your choices based on device characteristics and performance requirements.
8. A multimedia system processes audio and video streams while supporting user interaction. Design a comprehensive I/O strategy that combines all three transfer techniques. Explain how you would prioritize different types of I/O operations and manage system resources.
9. Explain how vectored interrupts differ from non-vectored interrupts with examples. Why are vectored interrupts more efficient?
10. Explain the working of a **synchronous bus** with the help of a timing diagram.
11. Differentiate between synchronous and asynchronous buses with suitable examples.
12. Draw and explain the block diagram of a serial interface circuit. How does it differ from a parallel interface circuit?
13. Illustrate the working of an asynchronous bus using handshaking signals. Why is it preferred in heterogeneous systems?
14. Consider a computer system that needs to transfer data between two subsystems of different speeds. Propose whether a **synchronous or asynchronous bus** is more suitable and justify your choice.
15. Evaluate the impact of **bus width and clock speed** on system performance. Which is more critical for synchronous buses?

MODULE3

1. Analyze the concept of locality of reference in computer systems. Explain temporal and spatial locality with suitable examples. How does locality of reference influence cache design decisions? Discuss the impact of poor locality on system performance.
2. a) Define locality of reference. Explain spatial and temporal locality with an example
b. Differentiate between cache write through and cache write back (3 marks)
3. What do you mean by a cache miss? Explain the steps to handle the situation where an instruction access results in a miss.
4. Which are the two cache writing mechanisms? Explain the advantages of them. Discuss the Principle of locality with a suitable example.
5. Define the terms in the context of cache memory: block, hit rate, miss rate, hit time, miss penalty. How a write buffer can be used to solve the problem associated with the write-through cache?
6. Explain any two cache mapping functions using suitable examples.

A computer system has a main memory consisting of 1M 16-bit words. It also has a 4K-word cache organized in the block-set-associative manner, with 4 blocks per set and 64 words per block. Calculate the number of bits in each of the Tag, Set, and Word fields.

7. Differentiate between direct mapped and fully associative cache. Explain the advantages and disadvantages of a set associative cache.
8. How do you relate set associative mapped cache with Direct mapped and associative mapped cache mechanisms?
Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the tag values? [Assume memory is byte addressable]
9. With a neat diagram, illustrate the basic organization of cache memory. Discuss how CPU interacts with cache and main memory.
10. Describe the different types of cache misses with examples. Explain the step-by-step process of handling a cache miss in a processor. What strategies can be employed to reduce each type of miss?
11. Design a cache system for a processor with the following requirements: 32KB cache size, 32-byte block size. Compare direct-mapped, 2-way set associative, and fully associative organizations for this cache. Calculate the number of sets, tag bits, and analyze the trade-offs in terms of hit rate, access time, and hardware complexity.
12. Design a three-level cache hierarchy for a high-performance processor. Justify your design choices for cache sizes, associativity, and block sizes at each level. Calculate the overall hit rate and average access time given: L1 (1ns, 90% hit), L2 (5ns, 85% hit), L3 (20ns, 95% hit), Main Memory (200ns). Analyze the performance benefits of this hierarchy.

13. With a neat diagram, illustrate the basic organization of cache memory. Discuss how CPU interacts with cache and main memory.
14. A CPU has a cache block size of 16 bytes and main memory of 64 KB. Show how address mapping is done for:
 - a) Direct mapping
 - b) Fully associative mapping
15. Demonstrate with an example how cache misses (compulsory, conflict, capacity) occur and suggest techniques to reduce them.
16. Explain how High Bandwidth Memory (HBM) provides higher throughput compared to conventional DRAM. Support your explanation with a neat block diagram. Understand the theory. Analyse the importance of HBM in AI workloads like deep learning training and inference. Provide examples.
17. Analyse the importance of HBM in AI workloads like deep learning training and inference. Provide examples. Apply the application.
18. Explain any four techniques for reducing cache misses with suitable examples.
19. Describe three software-level optimization techniques for reducing cache misses. Provide practical examples from matrix multiplication or image processing.
20. Analyze how increasing block size vs increasing associativity affects cache performance in terms of reducing conflict misses.